

Player Footstep Sounds Based On Surfaces

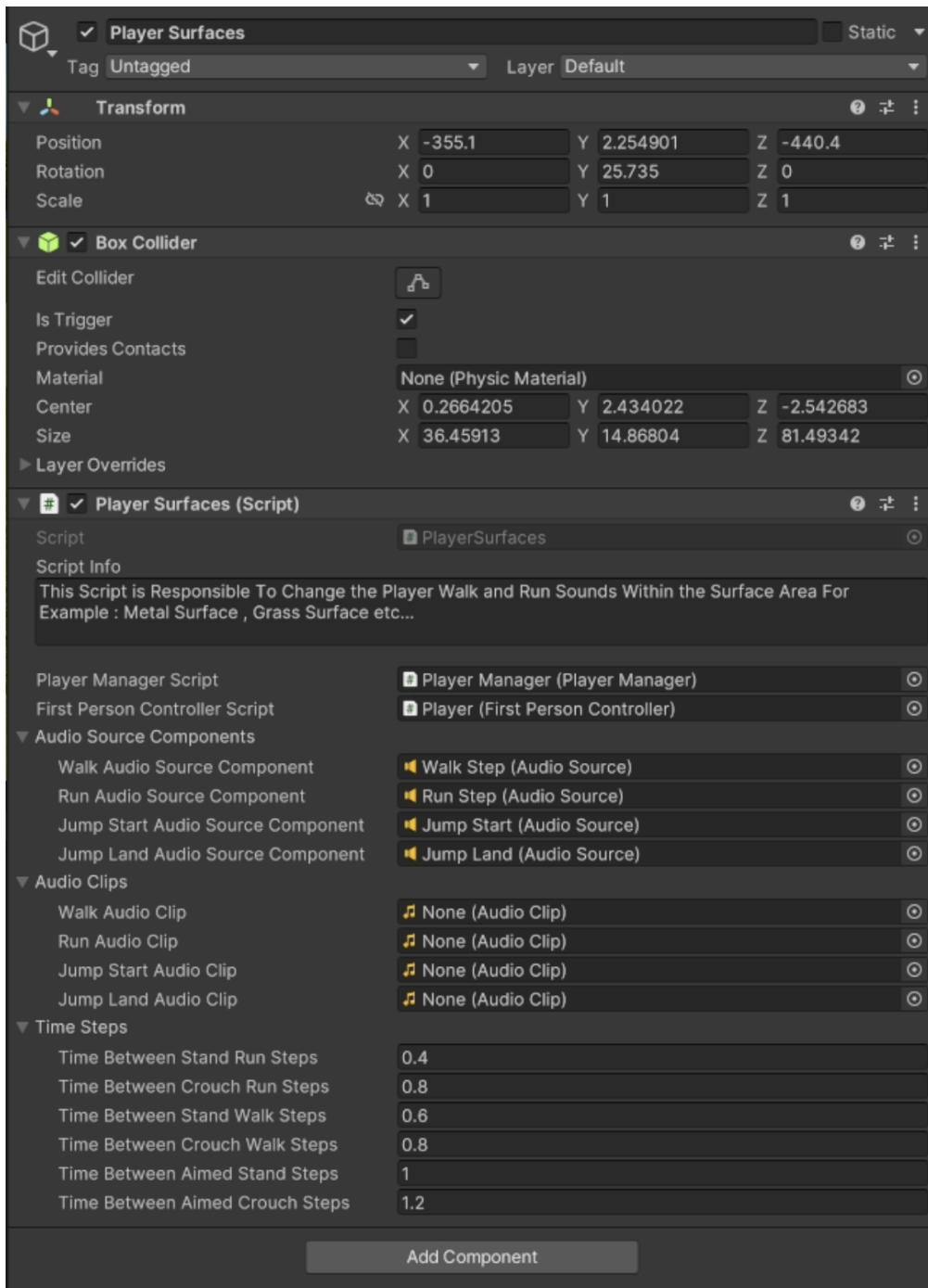
Introduction

This video will guide you on how to setup different footstep sounds at different surfaces i.e - Grass,Metal. There are two ways using which you can create footstep sounds at different surfaces. You can detect at which collider the player is moving by using Trigger based collider Or by using Raycast.

Player Surfaces (Collider Detection)

You can create any collider example: A Box collider and check its is Trigger Checkbox and add a component to that gameObject called 'PlayerSurfaces'. [See The Image below]

Script Info: This Script is Responsible To Change the Player Walk and Run Sounds Within the Surface Area For Example : Metal Surface , Grass Surface etc...



Fields	Info
PlayerManagerScript	Reference to the PlayerManager script for player-related functions.
FirstPersonControllerScript	Reference to the FirstPersonController

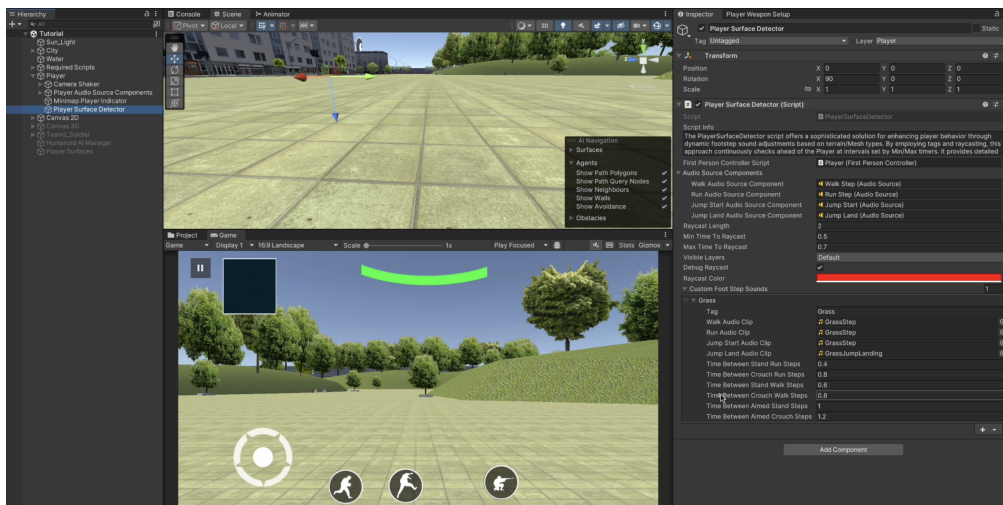
	script for movement-related functions.
AudioSourcesProperties.WalkAudioSourceComponent	Audio source for the walking sound.
AudioSourcesProperties.RunAudioSourceComponent	Audio source for the running sound.
AudioSourcesProperties.JumpStartAudioSourceComponent	Audio source for the jump start sound.
AudioSourcesProperties.JumpLandAudioSourceComponent	Audio source for the jump landing sound.
AudioClipProperties.WalkAudioClip	Audio clip to be played when walking.
AudioClipProperties.RunAudioClip	Audio clip to be played when running.
AudioClipProperties.JumpStartAudioClip	Audio clip to be played when starting a jump.
AudioClipProperties.JumpLandAudioClip	Audio clip to be played when landing after a jump.
TimeStepsProperties.TimeBetweenStandRunSteps	Time interval between each footstep sound while standing and running.
TimeStepsProperties.TimeBetweenCrouchRunSteps	Time interval between each footstep sound while crouching and running.
TimeStepsProperties.TimeBetweenStandWalkSteps	Time interval between each footstep sound

	while standing and walking.
TimeStepsProperties.TimeBetweenCrouchWalkSteps	Time interval between each footstep sound while crouching and walking.
TimeStepsProperties.TimeBetweenAimedStandSteps	Time interval between each footstep sound while aiming and standing.
TimeStepsProperties.TimeBetweenAimedCrouchSteps	Time interval between each footstep sound while aiming and crouching.
AudioSourceComponents	Contains references to the audio sources for different movement sounds.
AudioClips	Contains references to the audio clips for different movement sounds.
TimeSteps	Stores footstep time intervals based on different movement states.

Player Surface Detector (Raycast Detection)

You can create an empty gameObject and make it to be the child of 'Player' and rotate it's X Axis at 90 degree and position it little bit above from the ground and add the component to it called 'PlayerSurfaceDetector' and using the script you can tweak different values and decide which audio clip to playback when the player walk/run/jump on a certain surface with a unique tag.[See The Image below]

Script Info: The PlayerSurfaceDetector script offers a sophisticated solution for enhancing player behavior through dynamic footstep sound adjustments based on terrain/Mesh types. By employing tags and raycasting, this approach continuously checks ahead of the Player at intervals set by Min/Max timers. It provides detailed environmental responsiveness, allowing player to interact realistically with various surfaces like grass, concrete, or wood. However, due to its continuous raycasting nature, developers should consider performance implications, particularly on mobile platforms where it may slightly impact FPS. For smaller environments, alternative script like 'PlayerSurfaces' which do collider-based detection may be more performance-efficient, ensuring optimal gameplay experience across different device capabilities.



Fields	Info
FirstPersonControllerScript	Reference to the FirstPersonController script for player movement and footstep handling.
AudioSourceComponents	Holds references to different audio sources used for movement sounds.
raycastLength	Length of the raycast that detects surfaces in front of the Player. Adjust this value based on how far ahead you want the Player to detect surfaces.
minTimeToRaycast	Minimum time interval between consecutive raycasts. Controls how often the script performs raycast checks.

maxTimeToRaycast	Maximum time interval between consecutive raycasts. Controls how often the script performs raycast checks.
VisibleLayers	Visible layers during raycasting. Objects on these layers will trigger surface detections.
debugRaycast	Toggle to enable or disable visual debugging of raycasts in the Scene view. When enabled (true), the raycast is visualized with raycastColor during development and testing.
raycastColor	Color used to visualize the raycast in the Scene view. Helps developers see where the raycast is directed during debugging.
customFootStepSounds	List of custom footstep sounds for different surface tags. Populate this list with tags and corresponding audio clips to simulate realistic footstep behavior based on terrain type.
WalkAudioSourceComponent	Audio source for walking sounds.
RunAudioSourceComponent	Audio source for running sounds.
JumpStartAudioSourceComponent	Audio source for jump start sounds.
JumpLandAudioSourceComponent	Audio source for jump landing sounds.
WalkAudioClip	Audio clip for walking on this surface type.
RunAudioClip	Audio clip for running on this surface type.
JumpStartAudioClip	Audio clip for jump start on this surface type.
JumpLandAudioClip	Audio clip for jump landing on this surface type.
TimeBetweenStandRunSteps	Time interval between each footstep while standing and running on this surface.
TimeBetweenCrouchRunSteps	Time interval between each footstep while crouching and running on this surface.

TimeBetweenStandWalkSteps	Time interval between each footstep while standing and walking on this surface.
TimeBetweenCrouchWalkSteps	Time interval between each footstep while crouching and walking on this surface.
TimeBetweenAimedStandSteps	Time interval between each footstep while aiming and standing on this surface.
TimeBetweenAimedCrouchSteps	Time interval between each footstep while aiming and crouching on this surface.